

Official Research Paper

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Title: Database Management in Library Cataloging

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I. Statement of Purpose:

Libraries contain tens of thousands of physical books, CDs, and media. Some even manage their own digital libraries. Every item, be it a physical book or a digital copy, has a digital bibliographic record stored on the libraries' ILS (Integrated Library System). This record stores information about the book such as author, title, edition, description, location within the library, whether it has been checked out or on hold, and much more. Catalogers must manage and update a library's collection daily to provide the best public service. Catalogers must also ensure patrons can have access to all of the library's collection by consistently managing the Online Public Access Catalog (OPAC). In my project, I hope to understand how libraries manage such vast amounts of data. I also hope to research some ways to improve editing and updating the data in bulk.

II. Background:

A library catalog is a list of library materials contained in a collection and forms the basis for access to the library's collection and this is the primary way of organizing materials. Modern library catalogs are often called online public access catalogs (OPACs). Catalogs are needed for retrieval (because most collections are too large for someone to remember every item) and inventory (because catalogs serve as a record for what is owned and as a reminder of what has been acquired, lost, replaced). First catalogs were handwritten books, then cards (which were easier to change once a collection changes), then printed cards. In the 1960s, the Library of Congress developed the LC MARC format to automate the procedure of producing catalog cards. In the 1970s, computers began to organize catalog cards into a computerized card catalog. Not long after, the OPAC was developed.

Cataloging is the ongoing management of the discoverability of the library's collections. It is the process of creating metadata (information about information resources) to describe and organize library materials, making them accessible to users through a catalog. There are three parts of cataloging: Descriptive, Classification, and Subject Analysis.

Descriptive cataloging describes the item by notating the title, statement of responsibility (who was responsible for the intellectual content), edition information, publication information (who published the item, where it was published and the date of publication), various notes and access points. Descriptive cataloging is taking information from the item in hand (a physical item) or an online resource and describing it in such a way that the item can be identified and retrieved. To maintain consistency in descriptive cataloging, a set of rules was developed. One version of these rules is AACR2R (Anglo-American Cataloging Rules, 2nd edition, 2003 revision). Some libraries continue to use AACR2R, but many have changed to RDA (Resource Description and Access). RDA was built on the foundation of AACR2R and implemented by the Library of Congress in 2013. However, libraries still create records following AACR2R or RDA or records that are a hybrid of the rules. The rules are constantly being reviewed and revised but are extremely important to create a standard for descriptive cataloging and a reference / guide for what librarians are to do. Descriptive cataloging is especially crucial in adding access points to bibliographic records. Access points are key terms within the bibliographic record that make that item more searchable and can be pinpointed by an algorithm or computer. They include the ISBN for a book, the publisher number for a DVD, call numbers, titles, people associated with the item (the author and/or illustrator of a book, an actor in a movie, a musical group, the editor of a book, etc.), or series.

Classification is the systematic arrangement of books and other materials on shelves. The two most common classification systems used in the U.S. are the Dewey Decimal Classification (DDC) and the Library of Congress Classification System (LCC). Classification numbers can also be access points in the OPAC. Most libraries index the classification number, and if the number is indexed, it is searchable.

Subject analysis uses controlled vocabulary to tell what the item is about and/or to describe what the item is. Controlled vocabulary means that everyone is using the same term to describe the same concept or type of item. The two most common subject heading systems used in the U.S. are Library of Congress Subject Headings (LCSH) and Sears List of Subject Headings. The four types of subject headings include topical headings, named entities, bibliographic form, and literary form.

Why is this formatting necessary? A look into metadata, access and retrieval,

Metadata is commonly thought of as “data about data”, but more specifically in terms of library use, it is about how information is placed in a more precise semantic

context to enable it to be used in other applications. Some metadata are mediated by a system to fulfill system needs, while others are mediated to be read by a user. Catalogers record essentially the same type of information in two different ways: as it appears in the resource and as it should appear for the system to index it. The primary job of catalogers is to determine what type of criteria users are likely to provide in their search and how to ensure that users receive accurate results to their queries. In other words, the more specific and standard information is formatted, the easier that item is to search.

After the cataloger has decided what information goes into the bibliographic description and has assigned all the access points including the call number and the subject headings, all the information is entered into a bibliographic record. This bibliographic record forms the basis of the catalog. Historically, in an unautomated environment, bibliographic records take the form of catalog cards and are filed in a card catalog. In an automated environment, bibliographic records take the form of **MARC records**.

This research paper primarily surrounds the management of these records, a process otherwise known as database cleanup or bibliographic file maintenance (BFM).

How are these records created or acquired? Where is it stored?

The bibliographic records are stored on a libraries' ILS (Integrated Library System). The ILS functions far more than cataloging, but that is not a concern for this paper. Records for new books rarely need to be created from scratch, as they are frequently provided by the book vendor or already on the local database. If there are none, certain vendors sell these records for a small fee. The Library of Congress provides an OPAC where records can be copied for free. However, in some instances, the records do need to be created from scratch in a process called "Original Cataloging".

However, this research paper surrounds the bulk editing of certain fields within MARC Records rather than the whole of a record.

Anatomy of a MARC record/ Understanding the MARC encoding standard:

Computers cannot directly read bibliographic information. Catalogers must not only create descriptive content but also essentially hard code that content so it could be machine actionable. MARC is a common data format for representing bibliographic metadata. MARC is very similar to HTML or CSS in that it's learning a set of tags. MARC organizes information in a catalog using numbered fields (Field tags) each with two coded indicators and sometimes coded subfields. That tag number indicates the type of information in that field.

Initially MARC was an extraordinary innovation by permitting transmission of data between computers, giving a huge boost to cooperative cataloging efforts and ultimately became how library systems and databases could store, index, and display

catalog records in an online discovery interface. But now the rigid set of numerical fields and subfields constrain the richness and discoverability of descriptive data. RDA was complicated by the inability of MARC to accommodate the kind of granular data RDA encourages us to record.

What type of edits/ upkeep/ cleanup is required of this data?

Beyond just the deletion/ creation/ addition of a record, one common issue with bibliographic file maintenance (BFM), is the adjustment work required when an authority record changes the access point that should be used in associated bibliographic records. Libraries are notified of these changes to authority records through various channels such as add-on library system features or a vendor provided service. Another common source of data cleanup is batch loading records into the library system. This can be caused by a major change in cataloging conventions rendering old data obsolete (such as the retiring of general material designation or GMD, a note about the format of a resource which was routinely included in the 245 field under AACR2, and now which is removed under RDA). Along with deleting fields, there is also fixing common grammatical errors within text to be displayed on an OPAC and ensuring subject headings are in line with the Library of Congress standards.

Understanding an issue on the local level:

Librarians usually bulk edit MARC records in a record set of 50-200 items and need to scan each individual record. This process is tiring and prone to human error. However, there are many tools to aid librarians in this process. MarcEdit and OpenRefine are powerful editing tools for large batches of records. Some catalogers learn programming languages such as XSLT or Python.

Helpful Programming languages for Catalogers:

XSLT: a styling language for XML, transforms XML documents into formats such as HTML or PDF. In metadata work, XSLT is primarily used to transform XML metadata from one standard to another

Python: Python libraries are useful in working with library metadata, with a specific set of functions to use without writing one's own codes. (PyMarc is especially useful in editing MARC records)

GREL (General Refine Expression Language): GREL is a scripting language used specifically within OpenRefine for complex data transformations (however, OpenRefine also supports Python, so GREL is not a necessity)

III. My Project and Process:

Unfortunately, there is no set guide or course or class on applying OpenRefine to bibliographic data, so many librarians are left to figure it out on their own (despite many libraries lacking the time, funds, and resources to devote part of their day to experimenting). This is where my project began.

After learning the background and learning how to navigate Polaris (Glendale Public Libraries' ILS), I began my own independent research and experimenting with various tools such as OpenRefine, MarcEdit, and Excel to develop a more efficient process to bulk edit MARC records. Once I developed a successful process, I began writing a Standard Operating Procedure (SOP) detailing that process. Once the SOP was completed, it was tested and used by my supervisor who, after suggesting certain edits, approved it. I also wrote a smaller SOP for a different field.

I am proud to say that the SOPs are documented in the library and will be used in the future. The SOPs, alongside this research paper, will be part of my final product.

Detailing my results:

I focused specifically on editing the 520 and 650 fields (though the process could be applied to other fields). The 520 field in a MARC record is a summary, abstract, annotation, or review of the material (book, audiobook, CD, etc...). It can contain grammatical errors. The 650 fields (including 650, 655, etc...) are the Library of Congress headings (topical, genre etc...) which require a specific syntax and formatting mandated by the Library of Congress.

For the 520 fields, after experimenting with record sets of 200 and 80 items respectively, I developed a process using MarcEdit, OpenRefine, and Excel to automate fixing grammatical errors (all three needed to be used in tandem). For the 650 fields, after also experimenting with record sets of 200 and 80 items, the process only required MarcEdit and OpenRefine. However, it does require an understanding of the proper formatting for the Library of Congress subject headings.

To learn more about the specifics and code, reference my SOP.

IV. Analysis/ Difficulties

When entering this project, I was unclear on what I wanted to do or what I hoped to achieve because I was unfamiliar with the field, and this slowed down the process in the initial stages and first few weeks. Learning the background took longer than anticipated because I needed to understand the bigger picture before narrowing down the scope of my project. Therefore, I had less time for my actual independent research. I could not learn in depth a new programming language (GREL) in the span of two months, so I was limited in the data transformations I could perform and detail in the SOP (that being said, I did use the basics of GREL for certain complicated transformations). I

was also limited in learning the MARC encoding standards, which limited the type of edits and which fields I could work with.

V. Significance and Bigger Picture (Linked Data)

The whole reason for content standards to exist is to increase accessibility to all the materials for a libraries' patron. The better and more concise a material is described, the more discoverable it is. The better managed the database is, the better searchable all materials in the library are. It also ensures the most factual and concise information about a material is provided when browsing the catalog.

Furthermore, the adherence to the formats and standards mandated by the Library of Congress contributes to the larger movement standardizing records in library databases across the nation. Why should libraries strive to standardize their records? For Linked Data.

In general, linked data breaks down information silos and makes it easier to browse through complex data. The more linked data is out there, the closer we are to creating the 'Web of Data' – a global network of interconnected machine-readable information, as opposed to a vast collection of unconnected datasets. Specifically in terms of libraries, Library linked data (LLD) refers to the use of linked data standards applied to bibliographic and authority data sets, with the hope of decreasing redundant cataloging work and increasing visibility of library resources and interoperability with non-library systems.

In other words, the more standardized and uniformly structured records are across libraries' databases, the easier it is for computers to find connections between works, and the easier it is for a computer to retrieve this record. The more data is opened on the Web (Open Data), the more integrated sets of data will be connected to the Semantic Web (Linked Open Data). Within this context, libraries can complement their data by linking it to other, external data sources. "The promise of the semantic web is a dazzling one. By marking up information in standardized, highly structured formats like Resource Description Framework (RDF), we can allow computers to better "understand" the meaning of content, rather than simply matching on strings of text. This would allow web search engines to function more like relational databases, providing much more accurate search results" (Byrne and Goddard, 2010).

The implication of the semantic web provides benefits for both patrons and librarians. Patrons would have increased access to materials provided by their local libraries because of faster and more accurate searches. Librarians could easily compile and publish data for their patrons. Linked data would allow usage statistics, survey results, and other key information to be published in a standardized format, making it easier to analyze, share, and compare with external sources like government or university

data. It would also greatly improve system interoperability, enabling libraries and vendors to build a shared knowledge hub that links information from publishers, vendors, and other libraries. This could streamline core functions like cataloging, resource management, and search, while replacing the fragmented tools (such as the ILS) libraries currently rely on. Most excitingly, linked data could enable powerful, semantically rich search tools—possibly surpassing the capabilities of current search engines—leading to smarter, more relevant search results.

Of course, there are a lot of obstacles to reaching this point. There are privacy concerns, lack of trust in open systems, complex rights management, and the significant organizational effort required to shift from current metadata models. While some early adopters are leading the way, broader library engagement will depend on increased education, practical demonstrations of value, and support for overcoming non-technical barriers. Most of all, local libraries are dealing with lackluster resources, time, and funds, and cannot properly devote time to upkeeping their databases.

The movement continues nonetheless, and librarians remain hopeful, and the public should as well. Libraries are a vital part of the larger public good. They allow for affordable access to knowledge, digital literacy, community, and civic engagement. The better libraries are equipped to manage their own collections, the sooner those collections are available to the public.

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